

Jeongsoo Choi

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Research Interest

My research interests lie in Multimodal Machine Learning, Speech Processing, and applications.

- Multilingual and Multimodal Machine Translation
- Audio-Visual Speech Recognition and Generation

Work Experience

Microsoft , Beijing, China Research Intern, Speech Team - Advisors: Dr. Shujie Liu and Dr. Jinyu Li	May 2024 - Oct. 2024
Meta , Seattle, WA, USA Research Scientist Intern, FAIR Team - Advisor: Dr. Hongyu Gong	Jun. 2025 - Present

Education

Ph.D. in Electrical Engineering, KAIST - Advisor: Prof. Joon Son Chung	Sep. 2020 - Present
B.S. in Electrical Engineering, KAIST	Mar. 2015 - Aug. 2020

Publications

<C: Conference, J: Journal, P: Preprint, *: Equal Contribution>

[C14] **Accelerating Diffusion-based Text-to-Speech Model Training with Dual Modality Alignment**
Jeongsoo Choi*, Zhikang Niu*, Ji-Hoon Kim, Chunhui Wang, Joon Son Chung, Chen Xie
Interspeech 2025

[C13] **From Faces to Voices: Learning Hierarchical Representations for High-quality Video-to-Speech**
Ji-Hoon Kim, Jeongsoo Choi, Jaehun Kim, Chaeyoung Jung, Joon Son Chung
CVPR 2025 Highlight Presentation

[C12] **ARLON: Boosting Diffusion Transformers with Autoregressive Models for Long Video Generation**
Zongyi Li*, Shujie Hu*, Shujie Liu, Long Zhou, Jeongsoo Choi, Lingwei Meng, Xun Guo, Jinyu Li, Hefei Ling, Furu Wei
ICLR 2025

[C11] **V2SFlow: Video-to-Speech Generation with Speech Decomposition and Rectified Flow**
Jeongsoo Choi*, Ji-Hoon Kim*, Jinyu Li, Joon Son Chung, Shujie Liu
ICASSP 2025

- [C10] **Accelerating Codec-based Speech Synthesis with Multi-Token Prediction and Speculative Decoding**
Tan Dat Nguyen, Ji-Hoon Kim, Jeongsoo Choi, Shukjae Choi, Jinseok Park, Younglo Lee, Joon Son Chung
ICASSP 2025
- [P5] **Dub-S2ST: Textless Speech-to-Speech Translation for Seamless Dubbing**
Jeongsoo Choi*, Jaehun Kim*, Joon Son Chung
arXiv May 2025, *Under review*
- [P4] **AlignDiT: Multimodal Aligned Diffusion Transformer for Synchronized Speech Generation**
Jeongsoo Choi, Ji-Hoon Kim, Kim Sung-Bin, Tae-Hyun Oh, Joon Son Chung
arXiv Apr. 2025, *Under review*
- [P3] **VoiceCraft-Dub: Automated Video Dubbing with Neural Codec Language Models**
Kim Sung-Bin, Jeongsoo Choi, Puyuan Peng, Joon Son Chung, Tae-Hyun Oh, David Harwath
arXiv Apr. 2024, *Under review*
- [P2] **MAVFlow: Preserving Paralinguistic Elements with Conditional Flow Matching for Zero-Shot AV2AV Multilingual Translation**
Sungwoo Cho, Jeongsoo Choi, Sungnyun Kim, Se-Young Yun
arXiv Apr. 2025, *Under review*
- [P1] **Deep Understanding of Sign Language for Sign to Subtitle Alignment**
Youngjoon Jang*, Jeongsoo Choi*, Junseok Ahn, Joon Son Chung
arXiv Mar. 2025, *Under review*
- [J2] **Textless Unit-to-Unit training for Many-to-Many Multilingual Speech-to-Speech Translation**
Minsu Kim*, Jeongsoo Choi*, Dahun Kim, Yong Man Ro
IEEE/ACM TASLP (2024)
- [C9] **AV2AV: Direct Audio-Visual Speech to Audio-Visual Speech Translation with Unified Audio-Visual Speech Representation**
Jeongsoo Choi*, Se Jin Park*, Minsu Kim*, Yong Man Ro
CVPR 2024 Highlight Presentation
- [C8] **Text-driven Talking Face Synthesis by Reprogramming Audio-driven Models**
Jeongsoo Choi, Minsu Kim, Se Jin Park, Yong Man Ro
ICASSP 2024
- [C7] **Towards Practical and Efficient Image-to-Speech Captioning with Vision-Language Pre-training and Multi-modal Tokens**
Minsu Kim, Jeongsoo Choi, Soumi Maiti, Jeong Hun Yeo, Shinji Watanabe, Yong Man Ro
ICASSP 2024
- [C6] **Exploring Phonetic Context-Aware Lip-Sync For Talking Face Generation**
Se Jin Park, Minsu Kim, Jeongsoo Choi, Yong Man Ro
ICASSP 2024
- [J1] **AKVSR: Audio Knowledge Empowered Visual Speech Recognition by Compressing Audio**
Jeong Hun Yeo, Minsu Kim, Jeongsoo Choi, Dae Hoe Kim, Yong Man Ro
IEEE TMM (2024)

[C5] **DiffV2S: Diffusion-based Video-to-Speech Synthesis with Vision-guided Speaker Embedding**
Jeongsoo Choi*, Joanna Hong*, Yong Man Ro
ICCV 2023

[C4] **Lip Reading for Low-resource Languages by Learning and Combining General Speech Knowledge and Language-specific Knowledge**
Minsu Kim*, Jeong Hun Yeo*, Jeongsoo Choi, Yong Man Ro
ICCV 2023

[C3] **Intelligible Lip-to-Speech Synthesis with Speech Units**
Jeongsoo Choi, Minsu Kim, Yong Man Ro
Interspeech 2023

[C2] **Watch or Listen: Robust Audio-Visual Speech Recognition With Visual Corruption Modeling**
Joanna Hong*, Minsu Kim*, Jeongsoo Choi, Yong Man Ro
CVPR 2023

[C1] **SyncTalkFace: Talking Face Generation with Precise Lip-Syncing via Audio-Lip Memory**
Se Jin Park, Minsu Kim, Joanna Hong, Jeongsoo Choi, Yong Man Ro
AAAI 2022 Oral Presentation

Programming Skills

Languages

- Python, C/C++, and MATLAB

Deep Learning Libraries

- Pytorch, Fairseq, and HuggingFace

References

Prof. Joon Son Chung

- School of Electrical Engineering, KAIST
- Email: joonson@kaist.ac.kr
- Homepage: <https://mmai.io/joon>
- Relationship: Ph.D. advisor at KAIST